Machine Learning Laboratory
Experiment 08
Evaluation of Classification Models

Aim

To evaluate and compare the performance of a Decision Tree Classifier and a Gaussian Naive Bayes Classifier using Accuracy, Precision, Recall, F1-score, ROC Curve and AUC.

Objectives

After completing this experiment, students will be able to:

1. Train two different classification models on the same dataset.
2. Generate and interpret the Confusion Matrix.
3. Calculate Accuracy, Precision, Recall and F1-score.
4. Obtain prediction probabilities from classification models.
5. Plot and compare ROC curves.
6. Calculate and compare AUC values.
7. Understand how evaluation metrics help in selecting an appropriate classification model.

Problem Statement

A breast cancer dataset contains numerical measurements describing characteristics of breast tumour cells. Develop two classification models:

1. Decision Tree Classifier
2. Gaussian Naive Bayes Classifier

Use the models to classify the tumour into the two target classes.
Evaluate both models using:

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1-score

- ROC Curve
- AUC

Compare the results obtained from both models and justify the selection of an appropriate model based on the evaluation metrics.

Software Requirements

- Python 3.x
- Jupyter Notebook / Google Colab / VS Code
- NumPy
- Pandas
- Matplotlib
- Scikit-learn

Dataset

The **Breast Cancer Wisconsin dataset** available in Scikit-learn is used in this experiment.

The dataset can be loaded directly using:

```
from sklearn.datasets import load_breast_cancer

data = load_breast_cancer()
```

The dataset contains numerical features describing characteristics of breast tumour cells and a binary target variable.

Evaluation Metrics

The following metrics are used to evaluate and compare the two classification models.

1. Confusion Matrix

For binary classification, the confusion matrix consists of four values:

	Predicted 0	**Predicted 1**
Actual 0	TN	FP
Actual 1	FN	TP

- **TP:** Actual positive and predicted positive.
- **TN:** Actual negative and predicted negative.
- **FP:** Actual negative but predicted positive.
- **FN:** Actual positive but predicted negative.

2. Accuracy

Accuracy measures the proportion of correctly classified samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

3. Precision

Precision measures how many of the samples predicted as positive are actually positive.

$$Precision = \frac{TP}{TP + FP}$$

4. Recall

Recall measures how many of the actual positive samples are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

5. F1-score

F1-score is the harmonic mean of Precision and Recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

6. ROC Curve

The Receiver Operating Characteristic (ROC) curve represents the relationship between the True Positive Rate (TPR) and False Positive Rate (FPR) at different classification thresholds.

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

7. AUC

AUC stands for **Area Under the ROC Curve**.

It summarizes the ability of a classification model to distinguish between the two classes across different classification thresholds.

AUC values closer to 1 indicate stronger class discrimination.

Important:
Accuracy, Precision, Recall and F1-score are calculated using predicted class labels.
ROC and AUC are calculated using prediction probabilities or decision scores.

Procedure

1. Import the required libraries.
2. Load the Breast Cancer dataset.
3. Separate the input features and target variable.
4. Divide the dataset into training and testing sets.
5. Create a Decision Tree Classifier.
6. Create a Gaussian Naive Bayes Classifier.
7. Train both models using the same training data.
8. Generate predictions for the same test data.
9. Calculate the Confusion Matrix for both models.
10. Calculate Accuracy, Precision, Recall and F1-score.
11. Obtain prediction probabilities from both models.
12. Generate the ROC curve for each model.
13. Plot both ROC curves on the same graph.
14. Calculate the AUC for both models.
15. Compare all the evaluation metrics.
16. Justify the choice of an appropriate model for the given problem.

Program

Step 1: Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import (
confusion_matrix,
accuracy_score,
precision_score,
recall_score,
```

```
f1_score,
roc_curve,
roc_auc_score
)
```

Step 2: Load the Dataset

```
data = load_breast_cancer()

X = data.data
y = data.target

print("Number of samples:", X.shape[0])
print("Number of features:", X.shape[1])
```

Step 3: Split the Dataset

Use the same training and testing data for both models so that their performance can be compared fairly.

```
X_train, X_test, y_train, y_test = train_test_split(
X,
y,
test_size=0.2,
random_state=42
)
```

Step 4: Create the Classification Models

```
dt_model = DecisionTreeClassifier(
random_state=42
)

nb_model = GaussianNB()
```

Step 5: Train Both Models

```
dt_model.fit(X_train, y_train)

nb_model.fit(X_train, y_train)
```

Step 6: Generate Predictions

```
y_pred_dt = dt_model.predict(X_test)

y_pred_nb = nb_model.predict(X_test)
```

Evaluation of Decision Tree

Step 7: Confusion Matrix

```
cm_dt = confusion_matrix(y_test, y_pred_dt)

print("Decision Tree Confusion Matrix:")
print(cm_dt)
```

Step 8: Calculate Evaluation Metrics

```
accuracy_dt = accuracy_score(y_test, y_pred_dt)

precision_dt = precision_score(
y_test, y_pred_dt
)

recall_dt = recall_score(
y_test, y_pred_dt
)

f1_dt = f1_score(
y_test, y_pred_dt
)

print("Decision Tree")
print("Accuracy :", accuracy_dt)
print("Precision:", precision_dt)
print("Recall :", recall_dt)
print("F1-score :", f1_dt)
```

Evaluation of Gaussian Naive Bayes

Step 9: Confusion Matrix

```
cm_nb = confusion_matrix(y_test, y_pred_nb)

print("Naive Bayes Confusion Matrix:")
print(cm_nb)
```

Step 10: Calculate Evaluation Metrics

```
accuracy_nb = accuracy_score(y_test, y_pred_nb)

precision_nb = precision_score(
y_test, y_pred_nb
)
```

```
recall_nb = recall_score(
y_test, y_pred_nb
)

f1_nb = f1_score(
y_test, y_pred_nb
)

print("Gaussian Naive Bayes")
print("Accuracy :", accuracy_nb)
print("Precision:", precision_nb)
print("Recall :", recall_nb)
print("F1-score :", f1_nb)
```

ROC Curve and AUC Comparison

Step 11: Obtain Prediction Probabilities

ROC curves are generated using prediction scores or probabilities rather than only the predicted class labels.

```
y_prob_dt = dt_model.predict_proba(X_test)[:, 1]

y_prob_nb = nb_model.predict_proba(X_test)[:, 1]
```

Here, `[:, 1]` selects the probability of the positive class.

Step 12: Calculate ROC Values

```
fpr_dt, tpr_dt, thresholds_dt = roc_curve(
y_test, y_prob_dt
)

fpr_nb, tpr_nb, thresholds_nb = roc_curve(
y_test, y_prob_nb
)
```

Step 13: Calculate AUC

```
auc_dt = roc_auc_score(
y_test, y_prob_dt
)

auc_nb = roc_auc_score(
y_test, y_prob_nb
)

print("Decision Tree AUC:", auc_dt)
```

```
print("Naive Bayes AUC:", auc_nb)
```

Step 14: Plot and Compare ROC Curves

```
plt.figure(figsize=(7, 5))

plt.plot(
fpr_dt,
tpr_dt,
label="Decision Tree"
)

plt.plot(
fpr_nb,
tpr_nb,
label="Gaussian Naive Bayes"
)

plt.plot(
[0, 1],
[0, 1],
linestyle="--",
label="Random Classifier"
)

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")

plt.legend()
plt.grid()
plt.show()
```

Step 15: Create a Comparison Table

```
comparison = pd.DataFrame({
"Metric": [
"Accuracy",
"Precision",
"Recall",
"F1-score",
"AUC"
],
"Decision Tree": [
accuracy_dt,
precision_dt,
recall_dt,
f1_dt,
```

```
auc_dt
],
"Gaussian Naive Bayes": [
accuracy_nb,
precision_nb,
recall_nb,
f1_nb,
auc_nb
]
})

print(comparison)
```

Observation Table

Record the obtained values in the following table.

Metric	Decision Tree	Gaussian Naive Bayes
Accuracy		
Precision		
Recall		
F1-score		
AUC		

Model Comparison and Interpretation

The two models should be compared using the complete set of evaluation metrics.

Students should answer the following questions:

1. Which model has higher Accuracy?
2. Which model has higher Precision?
3. Which model has higher Recall?
4. Which model has higher F1-score?
5. Which model has higher AUC?
6. Do all the evaluation metrics identify the same model as better?
7. Which model would you select for this classification problem? Give a reason based on the evaluation metrics.

Important:
A model should not automatically be selected only because it has the highest Accuracy.
The appropriate metric depends on the objective of the classification problem. In applications such as medical diagnosis, Recall may receive particular attention because False Negatives can be important.

Result

The Decision Tree and Gaussian Naive Bayes classification models were successfully trained and evaluated.

Their performance was compared using the Confusion Matrix, Accuracy, Precision, Recall, F1-score, ROC Curve and AUC. Based on the obtained evaluation results, an appropriate classification model was identified for the given problem.

Questions for Practice

1. Why should both models be evaluated on the same test dataset?
2. Why is Accuracy alone not always sufficient for model evaluation?
3. What is the difference between Precision and Recall?
4. Why is F1-score called a balance between Precision and Recall?
5. Why are prediction probabilities required for ROC-AUC?
6. What does the ROC curve represent?
7. What do TPR and FPR represent?
8. What does a higher AUC indicate?
9. Can a model have higher Accuracy but lower Recall than another model?
10. Which metric would you consider important if False Negatives are costly?
11. Why can different evaluation metrics give different conclusions about two models?
12. Why should model selection depend on the application?

Viva Questions

1. What is a classification model?
2. What is a Confusion Matrix?
3. Define TP, TN, FP and FN.
4. Write the formula for Accuracy.
5. Write the formula for Precision.
6. Write the formula for Recall.
7. Write the formula for F1-score.
8. What is ROC?
9. What is AUC?

10. What is the difference between a predicted class and predicted probability?
11. Which function is used to obtain prediction probabilities in Scikit-learn?
12. Why is $\texttt{predict}_proba()used for ROC - AUC$?
12. What does an AUC close to 1 indicate?
13. What does the diagonal line in the ROC graph represent?
14. Name the two classification models used in this experiment.
15. Why is Gaussian Naive Bayes used for this dataset?